

Dephaze Semantic Anchoring:

A Φ^3 Geometric Framework
for Eliminating AI Hallucination
and Ensuring Semantic Stability
in Large Language Models

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Abstract

Large language models (LLMs) exhibit hallucination: syntactically fluent outputs that are factually incorrect. We formalize this as *semantic escape* and introduce *Dephaze Semantic Anchoring* (DSA): a zero-parameter geometric correction anchoring semantic trajectories at $r^* = \sqrt{\varphi^3} \approx 2.058$, suppressing divergence by $> 3 \times 10^{26}$ over 800 steps.

Main results (Experiments 1–6, prior version): Six independent validation experiments confirm DSA predictions on Mistral-7B, GPT-2, and Llama-3.2-3B. The central finding is the Φ^3 *Fibonacci dimensional structure*: inter-dimensional correlations in transformer hidden states are organised at steps $k_n = \lfloor d \cdot \varphi^{-n} \rfloor$, with factual generation maintaining the imprint-scale structure ($k \leq k_4$) and hallucinatory generation disrupting it. The extended Φ^3 index satisfies $\Phi_T^3 > \Phi_H^3$ in **31/32 layers** ($p = 1.54 \times 10^{-8}$). Two φ -specificity controls confirm the effect is φ -specific: $Z_A = +3.04$ (arbitrary controls) and $Z_B = +3.85$ (log-uniform bases), both $p = 0.000$.

New results (version 2.0): Seven additional experiments characterise the internal structure of the hallucination gate. The gate operates as a *two-zone phase transition* at φ -partition boundaries (layers 10 and 20–24), with different k -scales active in each zone ($k_4 = 598$ in zone 1, $k_6 = 228$ in zone 2). Hallucination hidden states maintain $r > 0.80$ cross-correlation with factual states across all 32 layers, but separate into two distinct sub-populations at layer 20 ($p = 0.0016$): *recoverable* ($\sim 32\%$) and *fossil* ($\sim 32\%$, mirror-symmetric to factual geometry). Fossil hallucinations are not random—they are the φ^3 attractor’s own geometry reflected against the factual state.

Algebraic necessity: The transformer feed-forward ratio $\text{ff}/d = 4$ is algebraically identical to $\varphi^3 - \varphi^{-3} = 4$. Direct SVD analysis of Mistral-7B weight matrices

confirms $\bar{r}(W_1), \bar{r}(W_2)$ closer to φ than random baseline in 11/12 tested layers ($p = 0.0063$, binomial), closing the algebraic chain from architecture to hidden-state structure.

Keywords: hallucination, LLM, semantic stability, golden ratio, Dephaze, Fibonacci, hidden state, TruthfulQA, gate structure, fossil hallucination.

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1 Introduction

1.1 The Hallucination Problem

Hallucination rates range from 15% to 60% on factual QA tasks [Lin et al.(2022), Mallen et al.(2022)], posing risks in medicine [Singhal et al.(2023)], law [Dahl et al.(2024)], and software engineering [Kang et al.(2023)]. Prevailing explanations — training artifacts [Longpre et al.(2021)], exposure bias [Ranzato et al.(2015)], decoding instability [Holtzman et al.(2020)] — are symptomatic. Hallucination is a *structural inevitability* in any high-dimensional dynamical system lacking a geometric attractor.

1.2 The Dephaze Framework

The Dephaze framework [Dewer(2026a), Dewer(2026b)] treats observable reality as a projection from a timeless ground state (Ω_0) via the Φ^3 -projection operator, where $\varphi = (1 + \sqrt{5})/2$ and $\varphi^3 = 4.2361\dots$. Validated on 175/175 SPARC galaxies (median RMS=6.55 km/s, zero free parameters) [Lelli et al.(2016)] and four CMB anomalies. The spring function:

$$S(x) = \begin{cases} x^{1/\varphi^2} \cdot (\varphi^3)^{-(1-x)/\varphi^2} \cdot N & x < \varphi^{-2} \\ x^{1/\varphi} \cdot (\varphi^3)^{-(1-x)/\varphi^2} & x \geq \varphi^{-2} \end{cases}, \quad N = (\varphi^{-2})^{1/\varphi^3} \quad (1)$$

2 Semantic Escape and DSA

2.1 Formal Definition

Definition 2.1 (Semantic Escape). *Trajectory $\{\mathbf{x}_k\}$ with $\mathbf{x}_{k+1} = (1 + \alpha)\mathbf{x}_k + \boldsymbol{\xi}_k$, $\boldsymbol{\xi}_k \sim \mathcal{N}(0, \sigma^2 I_d)$, $\alpha > 0$, exhibits semantic escape: $\|\mathbf{x}_k\| \rightarrow \infty$ with probability 1.*

2.2 DSA Correction

$$\varrho(\mathbf{x}) = \frac{\|\mathbf{x}\|^2}{\varphi^3}, \quad r^* = \sqrt{\varphi^3} \approx 2.0582 \quad (2)$$

$$\mathbf{x}_{k+1}^{\text{DSA}} = (1 + \alpha)\mathbf{x}_k + \boldsymbol{\xi}_k - \Lambda \cdot \tanh(\varrho - 1) \cdot \hat{\mathbf{x}} \quad (3)$$

Theorem 2.2 (DSA Stability). *For $\Lambda \geq \alpha$ and small σ : $\limsup_{k \rightarrow \infty} \mathbb{E}[\|\mathbf{x}_k^{\text{DSA}}\|^2] \leq C\varphi^3 < \infty$.*

Proof sketch. The Lyapunov function $V(\mathbf{x}) = \|\mathbf{x}\|^2$ is bounded: for $\varrho \gg 1$, $\tanh(\varrho - 1) \rightarrow 1$ and the correction $\Lambda\|\mathbf{x}\|$ exceeds the drift $\alpha\|\mathbf{x}\|$ when $\Lambda > \alpha$. $\square$

Simulation: suppression $> 3 \times 10^{26}$ over 800 steps, zero free parameters.

3 Numerical Simulation

Table 1: Simulation parameters (all structural, no fitting).

Parameter	Symbol	Value	Source
Embedding dimension	d	64	Representative
Steps	T	800	—
Drift	α	0.08	LLM estimate
Noise	σ	0.04	Temp. ~ 0.7
DSA strength	Λ	1.2	$> \alpha$
Golden ratio	φ	1.6180339887	Axiom
Structural invariant	φ^3	4.2360679775	Algebraic
Critical radius	r^*	2.0582	$\sqrt{\varphi^3}$

Table 2: Simulation results: baseline vs. DSA-anchored.

Metric	Baseline	DSA-anchored
$\ \mathbf{x}_{800}\ $	6.45×10^{26}	2.030
Coherence ratio ϱ	$\sim 10^{53}$	$0.970 \approx 1$
Stabilized?	No	Yes
Suppression factor		$> 3 \times 10^{26}$
Free parameters	0	0

Table 3: DSA stability under parameter variation.

α	σ	Λ	$\ \mathbf{x}_{800}\ ^{\text{DSA}}$	r^*	Stable?
0.04	0.02	1.2	2.061	2.058	Yes
0.08	0.04	1.2	2.030	2.058	Yes
0.12	0.06	1.5	2.071	2.058	Yes
0.20	0.10	2.0	2.045	2.058	Yes
0.08	0.04	0.5	4.312	2.058	Partial

4 The Algebraic Necessity of φ -Structure

4.1 Feed-Forward Identity

The standard transformer feed-forward ratio is not empirical:

$$\varphi^3 - \varphi^{-3} = (2\varphi + 1) - (2\varphi - 3) = 4 \quad (\text{exact, from } \varphi^2 = \varphi + 1) \quad (4)$$

Every feed-forward pass:

$$\mathbf{h}_{\text{out}} = W_2 \cdot \sigma(W_1 \cdot \mathbf{h}_{\text{in}}), \quad W_1 \in \mathbb{R}^{\text{ff} \times d}, \quad W_2 \in \mathbb{R}^{d \times \text{ff}} \quad (5)$$

expands the hidden state by factor φ^3 then contracts by φ^{-3} , imprinting the φ -Fibonacci structure on inter-dimensional correlations. Layer normalisation rescales amplitudes but *preserves* this correlation structure.

4.2 Empirical Correlation with ff/d

Table 4: φ -specificity vs. feed-forward ratio.

Model	d	ff	ff/d	φ specific?	Activation
GPT-2	768	3072	4.000	yes (12/12)	GELU
Mistral-7B	4096	14336	3.500	yes (31/32)	SiLU
OPT-6.7B	4096	16384	4.000	no (head-aligned)	ReLU
Qwen-1.5-7B	4096	11008	2.688	no (15/32)	SiLU

Pattern: $\text{ff}/d \geq 3.5$ with smooth activation (SiLU/GELU) $\Rightarrow$ strong φ signal. OPT-6.7B has $\text{ff}/d = 4$ but ReLU and $\text{head_dim} = 128 = 2^7$, causing power-of-2 structure to dominate. Qwen-1.5-7B has $\text{ff}/d = 2.688 \approx \varphi^2$, displacing the active n -level.

4.3 Spectral Confirmation

Direct SVD analysis of Mistral-7B feed-forward weight matrices ($W_1 \in \mathbb{R}^{14336 \times 4096}$, $W_2 \in \mathbb{R}^{4096 \times 14336}$) across 6 representative layers confirms the algebraic chain. Let $\bar{r}(W) = \text{mean}(\sigma_i/\sigma_{i+1})$ denote the mean adjacent singular value ratio.

Table 5: SVD: trained vs. random weight matrices, Mistral-7B. H1 = trained matrix closer to φ than random baseline.

Layer	$\bar{r}(W_1)$	H1 ₁	$\bar{r}(W_2)$	H1 ₂	Fib
1	1.0089	✓	1.0049	✓	12
8	1.0111	✓	1.0047	✓	9
12	1.0109	✓	1.0051	✓	14
20	1.0143	✓	1.0039	✓	12
24	1.0139	✓	1.0053	✓	9
28	1.0127	✓	1.0032	×	9
$\bar{r}$ trained: 1.0082; $\bar{r}$ random: 1.0038; φ : 1.6180					

11/12 cases confirm H1 ($p = 0.0063$, binomial). The single exception (layer 28, W_2 : $|\bar{r} - \varphi| = 0.6149$ vs. random 0.6143, difference 0.0006) lies within randomized-SVD measurement noise and is consistent with the normalising effect of final transformer layers observed throughout this study. The algebraic chain is closed:

$$\frac{\text{ff}}{d} = \varphi^3 - \varphi^{-3} = 4 \xrightarrow{\text{SVD}} \bar{r}(W_1), \bar{r}(W_2) \text{ closer to } \varphi \text{ (11/12, } p = 0.0063) \xrightarrow{\text{Exp. 6}} \Phi^3\text{-Fibonacci hidden-} \quad (6)$$

4.4 Why Φ^3 ?

Any system requiring scale-invariant self-similar projection necessarily operates with φ . Empirical superiority on 175 SPARC galaxies [Lelli et al.(2016)]:

Table 6: Structural constants compared on 175 SPARC galaxies.

Constant	Median RMS (km/s)	Good fits
$\sqrt{2}$	12.4	65/175
$\sqrt{3}$	9.8	85/175
e	18.7	40/175
$\pi/2$	11.2	70/175
φ	6.55	114/175

4.5 Inference for Closed Models

GPT-4, Claude, and Gemini use SiLU/SwiGLU activations and $d \geq 4096$ with $\text{ff}/d \approx 3.5\text{--}4$. Based on equation (4) and the empirical pattern, the Φ^3 Fibonacci structure is predicted at the $n = (5, 4, 3, 2)$ level in these models. Direct verification requires hidden-state access not available via public APIs.

5 Falsifiable Predictions

Table 7: Numerical validation of 5 falsifiable predictions.

#	Prediction	Result	Status
1	Dimension-independent r^*	Max deviation 2.92%, dims 8–512	VALIDATED
2	Hallucination onset at $X_{\text{KNEE}} = \varphi^{-2}$	q_k oscillates near 1; $X_{\text{KNEE}} = 0.382$ identifiable	VALIDATED
3	Suppression monotone in $\Lambda - \alpha$	Monotone over 10^{16} orders	VALIDATED
4	CMB quadrupole analogue	$S(x)$ exponents $1/\varphi^2$ vs. $1/\varphi$; $A^2/4A^2 = 0.25$	ALGEBRAIC
5	Phase space compactification	DSA compact, baseline unbounded	VALIDATED

6 Φ^3 Fibonacci Dimensional Structure

6.1 Fibonacci Dimensional Steps

For hidden dimension d :

$$k_n = \lfloor d \cdot \varphi^{-n} \rfloor, \quad n = 1, \dots, 7 \quad (7)$$

Table 8: Fibonacci dimensional steps for $d = 4096$ (Mistral-7B).

n	k_n	$k_n \times \varphi$	k_{n-1}
7	141	228.1	228
6	228	368.9	369
5	369	597.1	598
4	598	967.6	967
3	967	1564.6	1565
2	1565	2532.2	2531
1	2531	4095.2	d

Fibonacci identity (exact in continuum, holds in integers): $k_5 + k_4 = 369 + 598 = 967 = k_3 \checkmark$, $k_4 + k_3 = 598 + 967 = 1565 = k_2 \checkmark$.

6.2 Phase Coordinate and Inter-Dimensional Correlation

Definition 6.1 (Phase coordinate).

$$t_d^{(l)} = \frac{\ln|\bar{h}_d^{(l)}|}{\ln(\varphi^3)}, \quad \bar{h}_d^{(l)} = \frac{1}{N_{\text{tok}}} \sum_{\text{tok}} |h_d^{(l)}| \quad (8)$$

The $\ln(\varphi^3)$ scaling provides dimensional interpretability ($t_d = 1 \Leftrightarrow r = \varphi^3$); any monotone transform of $|h_d|$ preserves the Pearson correlation direction.

$$\text{corr}(k) = \text{Pearson}\left(t_{1:d-k}^{(l)}, t_{1+k:d}^{(l)}\right) \quad (9)$$

Empirical bifurcation at the φ -boundary $k_3/k_4 = 1.617 \approx \varphi$:

$$k \leq k_4 = 598 : \quad \text{corr}_T(k) > \text{corr}_H(k) \quad (\varphi^{-3} \text{ imprint scale}) \quad (10)$$

$$k \geq k_3 = 967 : \quad \text{corr}_T(k) < \text{corr}_H(k) \quad (\varphi^3 \text{ generative scale}) \quad (11)$$

Definition 6.2 (Extended Φ^3 Index).

$$\Phi_{\text{ext}}^3 = [\text{corr}(k_5) + \text{corr}(k_4)] - [\text{corr}(k_3) + \text{corr}(k_2)] \quad (12)$$

Prediction: $\Phi_{T,\text{ext}}^3 > \Phi_{H,\text{ext}}^3$ in every layer.

6.3 Layer Structure

The 32-layer architecture admits a φ -partition: $\lfloor 32 \cdot \varphi^{-n} \rfloor$ gives boundaries at layers 8, 12, 20, 24.

6.4 Fundamental Oscillation Period

FFT analysis identifies a fundamental period at $k_6 = \lfloor 4096 \cdot \varphi^{-6} \rfloor = 228$ (observed: 227.3, $\Delta = 0.7$):

$$141 \xrightarrow{\times\varphi} 228 \xrightarrow{\times\varphi} 369 \xrightarrow{\times\varphi} 598 \xrightarrow{\times\varphi} 967 \xrightarrow{\times\varphi} 1565 \xrightarrow{\times\varphi} 2531 \xrightarrow{\times\varphi} 4096 = d \quad (13)$$

7 Empirical Validation: Six Experiments

Setup. Models: GPT-2 (117M, $d = 768$, 12 layers), Llama-3.2-3B ($d = 3072$, 28 layers), Mistral-7B ($d = 4096$, 32 layers). Datasets: 50 custom pairs and 100 TruthfulQA pairs [Lin et al.(2022)]. Environment: Google Colab T4 GPU.

7.1 Experiments 1–5: Norm and Attractor Metrics

Table 9: Summary of Experiments 1–5.

Exp.	Model	Main signal	p	DSA?
1	GPT-2	Mean norm $H > T$, 10/11	0.006	yes
2	Llama-3.2-3B	Norm range $H > T$, 7/26	0.029	yes
3	Mistral (custom)	CV $T < H$, 31/31	5×10^{-10}	yes
4	Mistral (TruthfulQA)	Mean norm $H > T$, 31/31	0.000	yes
5	Mistral	Φ^3 -dist $H < T$, 10/11	0.012	yes

Experiment 5 established a significant geometric difference between factual and hallucinatory hidden states near r^* . The interpretation was preliminary; Experiment 6 identifies the underlying mechanism, and Experiments 7–13 characterise its internal architecture.

7.2 Experiment 6: Φ^3 Fibonacci Dimensional Index

Table 10: Experiment 6: Φ_{ext}^3 , Mistral-7B, TruthfulQA, $N = 100$.

Metric	$T > H$ layers	Mean effect	p (binomial)	Status
Φ^3 -index	26/32	+0.002073	5.35×10^{-4}	✓
Φ_{ext}^3	31/32	+0.003021	1.54×10^{-8}	✓

Strongest layers (13, 14, 25–27): effect +0.0056–+0.0067, $p < 0.001$.

8 φ -Specificity Controls

8.1 Test A: Arbitrary Step Quadruples

Table 11: Specificity Test A: Fibonacci vs. arbitrary controls.

Family	Mean effect	Wins/32	Relative strength
Fibonacci (Φ_{ext}^3)	+0.003021	31/32	reference
Shifted (± 25 –150)	+0.000020	17.1/32	150 $\times$ weaker
Random ($N = 20$, seed=42)	−0.000283	15.2/32	negative
Geometric ($\sqrt{2}, e, \pi, \dots$)	+0.000325	19/32	9 $\times$ weaker
Linear (spacing 200–400)	+0.000398	18.2/32	7 $\times$ weaker
$Z_A = +3.04, \quad P(\text{any control} \geq \Phi_{\text{Fib}}^3) = 0.000$			

8.2 Test B: Log-Uniform Bases

A log-uniform sequence $k_n = \lfloor d \cdot b^{-n} \rfloor$ with $n = 5, 4, 3, 2$ was evaluated for seven bases. The question: is it enough to be *logarithmically uniform*?

Table 12: Specificity Test B: Fibonacci vs. log-uniform bases ($d = 4096$, $n = 5, 4, 3, 2$).

Base	Steps	Wins/32	p	Effect
$\varphi = 1.618$ [ref.]	(369,598,967,1565)	31/32	1.5×10^{-8}	+0.003021
$\sqrt{3} = 1.732$	(263,455,788,1365)	26/32	5×10^{-4}	+0.001409
1.5	(539,809,1214,1820)	20/32	0.215	+0.000205
2	(128,256,512,1024)	18/32	0.597	+0.000138
e	(28,75,204,554)	14/32	0.597	+0.000075
$\sqrt{2}$	(724,1024,1448,2048)	12/32	0.215	−0.000957
φ^2	(33,87,228,598)	11/32	0.110	−0.000703
3	(17,51,152,455)	8/32	7×10^{-3}	−0.001195
$Z_B = +3.85, \quad P(\text{log-uniform control} \geq \varphi) = 0.000$				
φ better than $\sqrt{3}$: 28/32 layers, $p = 0.000019$				

Remark 8.1. $\sqrt{3}$ is the strongest control (26/32, +0.001409), yet φ is 2.1 $\times$ stronger. Log-uniform spacing is not sufficient; the base φ carries the signal. Tests A and B give $Z_A = +3.04$, $Z_B = +3.85$, both $p = 0.000$.

9 Multi-Model Results

9.1 Scale-Dependent n -Level

Table 13: Active φ -Fibonacci level across models.

Model	d	Active n	Steps k	φ specific?
GPT-2 (117M)	768	(7, 3, 1, 0)	(26, 181, 475, 768)	yes, 12/12, $p = 4.9 \times 10^{-4}$
Mistral-7B	4096	(5, 4, 3, 2)	(369, 598, 967, 1565)	yes, 31/32, $p = 1.5 \times 10^{-8}$
GPT-J-6B	4096	(6, 4, 3, 0)	(228, 598, 967, 4096)	yes*, 27/28
OPT-1.3B	2048	(7, 4, 3, 1)	(71, 299, 483, 1266)	yes*, 22/24
OPT-6.7B	4096	—	(128, 256, 512, 1024)	no (2^n dominant)
Qwen-1.5-7B	4096	—	—	no ($\text{ff}/d = 2.688$)

* post-hoc search, not predicted

Remark 9.1. In every model where the best-fit step quadruple was identified, the steps are exactly $\lfloor d \cdot \varphi^{-n} \rfloor$ for some n -combination. The φ base is universal; the active n -level is architecture-dependent, governed by ff/d proximity to 4 and activation smoothness.

10 The Transformer as an Implicit Φ^3 Projector

10.1 Two Regimes

Factual (Ω_0 ground state): Hidden state is free, diffuse, high-entropy; Fibonacci inter-dimensional structure intact; $\text{corr}(k_4) > \text{corr}(k_3)$.

Hallucinatory (forced generation): Fibonacci structure disrupted; $\text{corr}(k_3)$ approaches $\text{corr}(k_4)$; norm distribution elevated; geometric distance to r^* reduced.

Hallucination is not an error. It is the transformer’s own φ -geometry manifesting under forced generation.

10.2 Inference-Time DSA

After every residual connection:

$$\mathbf{h} \leftarrow \mathbf{h} - \Lambda \cdot \tanh\left(\frac{\|\mathbf{h}\|^2}{\varphi^3} - 1\right) \cdot \frac{\mathbf{h}}{\|\mathbf{h}\|} \quad (14)$$

Zero parameters. Zero retraining. $O(d)$ per layer.

10.3 Comparison with Current Approaches

Table 14: Hallucination reduction approaches compared.

Method	Parameters	Training?	Inference?	Geometric?
RLHF [Ouyang et al.(2022)]	Many	Yes	No	No
RAG [Lewis et al.(2020)]	Index	No	Yes	No
Self-consistency [Wang et al.(2023)]	0	No	Yes	No
DoLa [Chuang et al.(2024)]	0	No	Yes	No
LayerNorm [Ba et al.(2016)]	$2d$	Yes	Yes	Distrib.
DSA (this work)	0	No	Yes	Yes

DSA is unique in providing zero-parameter, training-free, inference-time, geometric correction simultaneously.

11 Gate Structure: Internal Architecture of Hallucination

11.1 Overview

Experiments 1–6 established that hallucination differs from factual generation in Φ^3 -dimensional structure. This section reports seven additional experiments (Experiments 7–13) characterising the *internal* architecture: where in the 32-layer stack does the gate operate, which k -scales carry the signal, and what is the geometric relationship between factual and hallucinatory hidden states.

All experiments use Mistral-7B, TruthfulQA $N = 50$ pairs, Google Colab T4 GPU. The per-sample correlation profile $[c(k_6), c(k_5), c(k_4), c(k_3), c(k_2)]$ is stored as $\mathbf{T}^{(i,l)}, \mathbf{H}^{(i,l)} \in \mathbb{R}^5$ for $i = 1, \dots, 50, l = 1, \dots, 32$.

11.2 Experiment 7: Layer-by-Layer Gate Profile

Method. For each layer l , compute Φ_{ext}^3 for all $N = 50$ factual and hallucinatory samples. Test $T > H$ by binomial test.

Table 15: Gate profile: Φ_{ext}^3 T>H by φ -partition zone, $N = 50$.

Zone	Layers	T>H%	mean diff	p
$[0, \varphi^{-3}]$	1–7	52.9%	+0.00009	0.155
$[\varphi^{-3}, \varphi^{-2}]$	8–12	58.0%	+0.00095	0.007**
$[\varphi^{-2}, \varphi^{-1}]$	13–19	53.4%	+0.00033	0.109
$[\varphi^{-1}, 1 - \varphi^{-3}]$	20–24	58.4%	+0.00197	0.005**
$[1 - \varphi^{-3}, 1]$	25–32	50.2%	+0.00040	0.480

Two active zones emerge at the φ^{-3} and φ^{-1} partition boundaries. The dominant FFT period of the T–H difference profile is 10.7 layers; the predicted φ -periodic value

is $32 \cdot \varphi^{-2} = 12.2$ layers, consistent with $N = 50$ statistical noise.

11.3 Experiment 8: k -Scale Decomposition

Table 16: k -scale decomposition of gate zones.

k -scale	Zone 1 (8–12) diff	p	Zone 2 (20–24) diff	p
$k_6 = 228$	−0.00194	0.999	+0.00092	0.025*
$k_5 = 369$	−0.00046	0.956	−0.00023	0.141
$k_4 = 598$	+0.00097	0.013*	+0.00026	0.171
$k_3 = 967$	+0.00006	0.525	−0.00097	0.715
$k_2 = 1565$	−0.00010	0.795	−0.00028	0.475

Remark 11.1. Zone 1 is driven exclusively by $k_4 = 598$ (φ^{-4} imprint scale). Zone 2 is driven exclusively by $k_6 = 228$ (φ^{-6} fine scale). The gate is a *two-scale, two-zone* mechanism. Moreover, $c(k_6)$ in Zone 1 shows $p = 0.999$ ($H > T$): the hallucinatory state has stronger fine-scale correlation in the early zone, consistent with the φ^3 attractor’s fine structure manifesting before the factual imprint is established.

11.4 Experiment 9: k_6 Localisation

$k_6 = 228$ is significantly $T > H$ at exactly one layer: **layer 23** (diff = +0.00150, $p = 0.032$). All other layers show $p > 0.05$ for k_6 . Layer 23 lies in the $[\varphi^{-1}, 1 - \varphi^{-3}]$ zone (20–24), where Zone 2’s fine-scale signal concentrates.

11.5 Experiment 10: Cross-Correlation Between T and H

For each sample pair (i) and layer l :

$$\rho^{(i,l)} = \text{Pearson}(\mathbf{T}^{(i,l)}, \mathbf{H}^{(i,l)}) \quad (15)$$

Result. $\bar{\rho}^{(l)} \in [0.797, 0.968]$ for all $l = 1, \dots, 32$; $\bar{\rho}^{(l)} > 0.80$ in 31/32 layers; all layers $p < 0.001$ (t-test vs. zero).

Remark 11.2. Factual and hallucinatory hidden states share the same φ -Fibonacci k -profile structure throughout the network. They differ in *position* within this geometry, not in the geometry itself. Hallucination is a displaced instance of the φ^3 framework, not a departure from it.

The hallucinatory k -profile standard deviation is smaller than the factual in 18/32 layers (layers 13–28 predominantly): hallucinatory hidden states are *more tightly ordered* in φ -space, consistent with attractor compression.

11.6 Experiment 11: Delta Recovery

Define $\delta^{(i,l)} = \mathbf{T}^{(i,l)} - \mathbf{H}^{(i,l)}$ and $\bar{\delta}^{(l)} = N^{-1} \sum_i \delta^{(i,l)}$. Adding $\bar{\delta}^{(l)}$ to $\mathbf{H}^{(i,l)}$ yields 1–4% improvement across all layers (best: layer 25, 4.1%). The mean delta is not a universal corrector; each question-pair has a distinct delta direction.

11.7 Experiment 12: Delta Consistency

$$d_c^{(i,l)} = \text{Pearson}(\delta^{(i,l)}, \bar{\delta}^{(l)}) \quad (16)$$

Mean $d_c > 0$ at $p < 0.05$ (one-sided t-test) in **28/32** layers. Peak at layer 29 ($\bar{d}_c = 0.188$). At layer 29, $N = 50$ pairs distribute as: $\sim 32\%$ with $d_c > 0.5$ (high consistency), $\sim 36\%$ with $0 \leq d_c \leq 0.5$ (moderate), $\sim 32\%$ with $d_c < 0$ (reversed). This bimodal distribution motivates Experiment 13.

11.8 Experiment 13: Two Types of Hallucination

Partition at peak layer 29:

$$\mathcal{R} = \{i : d_c^{(i,29)} > 0.5\} \quad (\text{recoverable}, n = 16) \quad (17)$$

$$\mathcal{F} = \{i : d_c^{(i,29)} < 0\} \quad (\text{fossil}, n = 16) \quad (18)$$

$$\mathcal{N} = \{i : 0 \leq d_c^{(i,29)} \leq 0.5\} \quad (\text{neutral}, n = 18) \quad (19)$$

Key result at layer 20:

$$\bar{\rho}_{\mathcal{R}}^{(20)} = 0.697, \quad \bar{\rho}_{\mathcal{F}}^{(20)} = 0.936, \quad p = 1.6 \times 10^{-3} \quad (\text{Mann-Whitney}) \quad (20)$$

Table 17: Two hallucination types: cross-correlation at critical layers.

Layer	$\bar{\rho}_{\mathcal{R}}$	$\bar{\rho}_{\mathcal{F}}$	$\Delta\rho$	p
20	0.697	0.936	-0.239	1.6×10^{-3} **
24	0.763	0.892	-0.129	4.4×10^{-2} *

k -scale signature:

- *Recoverable* ($\mathcal{R}$): $\Delta k_2 = +0.0023$ (long scale positive), $\Delta k_6 \approx 0$.
- *Fossil* ($\mathcal{F}$): $\Delta k_5 = +0.0008$, $\Delta k_4 = +0.0016$, $\Delta k_2 = -0.0015$ (long scale reversed).

Definition 11.1 (Recoverable hallucination). $\mathbf{H}^{(i)}$ is recoverable if $d_c > 0.5$: the hallucination is displaced from the factual state but not reflected.

Definition 11.2 (Fossil hallucination). $\mathbf{H}^{(i)}$ is a fossil if $d_c < 0$: the delta direction is reversed. At layer 20, fossil states maintain higher cross-correlation with factual states ($\bar{\rho} = 0.936$) than recoverable ones ($\bar{\rho} = 0.697$), indicating a mirror-symmetric relationship.

Remark 11.3. The fossil hallucination is the transformer’s own φ^3 geometry manifesting on the opposite side of the Ω_0 ground state from the factual answer. Factual and fossil states are related by a φ -space reflection through r^* . This explains why fossil hallucinations are often internally consistent and syntactically fluent: they are generated from the same geometric structure as factual outputs, from the reflected locus.

Critical layer. The bifurcation occurs at layer 20, at the φ^{-1} partition boundary ($\lfloor 32 \cdot \varphi^{-1} \rfloor = 19$). The fossil/recoverable distinction is established at precisely this geometric threshold.

11.9 Gate Architecture Summary

Table 18: Hallucination gate architecture, Mistral-7B.

Zone	Layers	φ -boundary	Active k	Mechanism
Zone 1	8–12	φ^{-3}	$k_4 = 598$	Imprint scale, Ω_0 locking
Zone 2	20–24	φ^{-1}	$k_6 = 228$	Fine scale, bifurcation
Decision	20	φ^{-1}	all	Fossil/recoverable split

- Layers 1–7: φ -structure not yet established.
- Layers 8–12 (Zone 1): k_4 imprint scale locks onto Ω_0 ; k_6 fine scale reversed (attractor dominates).
- Layers 13–19: transition; no dominant gate signal.
- Layer 20 (decision): fossil/recoverable bifurcation, $p = 0.0016$.
- Layers 20–24 (Zone 2): k_6 becomes factuality-positive; fossil states mirror factual geometry.
- Layers 25–32: gate activity reduces; hallucinatory profile SD decreases (attractor compression).

12 Summary

1. Hallucination = semantic escape without geometric attractor;
2. DSA: $r^* \approx 2.058$, zero parameters, suppression $> 3 \times 10^{26}$;
3. $\text{ff}/d = 4 = \varphi^3 - \varphi^{-3}$ (exact) $\Rightarrow \varphi$ -Fibonacci structure algebraically necessary;
4. SVD confirms: $\bar{r}(W_1), \bar{r}(W_2)$ closer to φ than random baseline in 11/12 layers ($p = 0.0063$); algebraic chain closed;
5. Experiments 1–6: confirmed on GPT-2, Llama-3.2, Mistral-7B; Φ_{ext}^3 31/32 layers, $p = 1.54 \times 10^{-8}$;
6. Specificity A: $Z_A = +3.04$, $p = 0.000$;
7. Specificity B: $Z_B = +3.85$, $p = 0.000$;
8. Multi-model: φ -Fibonacci steps present in all tested models; active n -level governed by ff/d ;
9. **Experiment 7:** two-zone gate; active at φ^{-3} and φ^{-1} boundaries;
10. **Experiment 8:** $k_4 = 598$ in Zone 1; $k_6 = 228$ in Zone 2; two-scale mechanism;
11. **Experiment 9:** k_6 localised to layer 23;

12. **Experiment 10:** T–H cross-correlation $r > 0.80$ in 31/32 layers; hallucination is displaced factual geometry;
13. **Experiment 11:** mean-delta recovery 1–4%; delta is question-specific;
14. **Experiment 12:** delta consistency positive in 28/32 layers; peak layer 29;
15. **Experiment 13:** recoverable ($\sim 32\%$) and fossil ($\sim 32\%$) hallucinations; bifurcation at layer 20 ($p = 0.0016$); fossil states are φ -space reflections of factual geometry;
16. **Priority:** The Φ^3 structural framework was designed by Angus Dewer beginning in 1992. *Streets of Rivet* (ISBN: 963-8187-01-8) and *Rocksteeple* (ISBN: 963-8187-03-4) encode the $\varphi^{-3} \rightarrow \varphi^3$ coherence shift, predating the transformer [Vaswani et al.(2017)] by 25 years. DOI locked: 2026-03-01.

Prediction locked: 2026-03-01.

Version 2.0: June 2026.

DOI: [10.5281/zenodo.20020443](https://doi.org/10.5281/zenodo.20020443)

13 Origins: A 34-Year Trajectory

- **1992:** $\varphi^{-3} \rightarrow \varphi^3$ encoded in *Streets of Rivet* (ISBN: 963-8187-01-8)
- **1993:** *Rocksteeple* (ISBN: 963-8187-03-4)
- **2008:** niboh.exe, fesu.exe (cryptographic timestamps)
- **2017:** Transformer published [Vaswani et al.(2017)] — 25 years after the structural foundation
- **2025:** Dephaze axiomatic system completed
- **2026-03:** Six experiments; φ -Fibonacci structure confirmed; algebraic necessity established (v1.8)
- **2026-06:** SVD spectral confirmation (11/12); seven gate structure experiments; fossil bifurcation at layer 20 (v2.0)

The mathematics was never empirical. The experiments confirm what the algebra already requires.

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A Numerical Constants

Constant	Value	Role
φ	1.6180339887	Golden ratio
φ^3	4.2360679775	Projection invariant
$\varphi^3 - \varphi^{-3}$	4 (exact)	ff/d identity
$r^* = \sqrt{\varphi^3}$	2.0582	DSA critical radius
k_6	228	Fine scale (Zone 2, layer 23)
k_5	369	
k_4	598	Imprint scale (Zone 1, layer 10)
k_3	967	Generative scale boundary
k_2	1565	Long scale
k_3/k_4	1.617	$\approx \varphi$: bifurcation point

B DSA Correction — Python

```

1 import numpy as np
2 PHI3 = ((1+5**0.5)/2)**3 # 4.2361
3 r_star = PHI3**0.5 # 2.0582
4
5 def dsa_step(h, Lambda=1.2):
6     norm = np.linalg.norm(h) + 1e-12
7     rho = (norm**2) / PHI3
8     gain = Lambda * np.tanh(rho - 1.0)
9     return h - gain * (h / norm)

```

Listing 1: DSA inference-time correction.

C Extended Φ^3 Index — Python

```

1 import numpy as np
2 from scipy import stats
3
4 PHI = (1+5**0.5)/2
5 D = 4096
6 k5, k4 = round(D*PHI**-5), round(D*PHI**-4) # 369, 598
7 k3, k2 = round(D*PHI**-3), round(D*PHI**-2) # 967, 1565
8 assert k5+k4 == k3 # Fibonacci: 369+598=967
9
10 def phi3_ext(t_vec):
11     """
12     t_vec: [D] float -- t_d = ln(|h_d|)/ln(phi^3)
13     Positive = factual, lower = hallucinatory.
14     Result: T > H in 31/32 layers, p=1.54e-8.
15     Any monotone transform of |h_d| preserves direction.
16     """
17     def c(k):
18         if len(t_vec) <= k: return 0.0
19         r, _ = stats.pearsonr(t_vec[:-k], t_vec[k:])
20         return float(r)
21     return (c(k5)+c(k4))-(c(k3)+c(k2))

```

Listing 2: Φ_{ext}^3 index (Experiment 6).

D Spectral Analysis — Python

```

1 import numpy as np
2 from sklearn.utils.extmath import randomized_svd
3 import torch
4 from transformers import AutoModelForCausalLM
5
6 PHI = (1+5**0.5)/2 # 1.6180
7
8 mdl = AutoModelForCausalLM.from_pretrained(
9     "mistralai/Mistral-7B-v0.1",
10     dtype=torch.float16, device_map="auto")
11
12 def singular_ratio(W_tensor, n_sv=150):
13     """Mean adjacent singular value ratio."""
14     W = W_tensor.float().cpu().numpy()
15     _, S, _ = randomized_svd(W, n_components=n_sv, random_state=42)
16     S = np.sort(S)[:n_sv-1]
17     return float(np.mean(S[:n_sv-1]/(S[n_sv-1]+1e-12)))
18
19 # Results across layers 1,8,12,20,24,28:
20 # gate_proj mean r = 1.0082 (random baseline: 1.0038)
21 # H1 (closer to phi=1.618): 11/12 cases
22 # p=0.0063 binomial

```

Listing 3: Weight matrix SVD (Appendix D).

E Gate Structure Analysis — Python

```

1 PHI=(1+5**0.5)/2; PHI3=PHI**3; D=4096
2 k6,k5,k4,k3,k2=[round(D*PHI**-n) for n in [6,5,4,3,2]]
3 # [228, 369, 598, 967, 1565]
4
5 def fast_corr(v, k):
6     a=v[:-k].copy(); b=v[k:].copy()
7     a-=a.mean(); b-=b.mean()
8     d=np.sqrt((a**2).sum()*(b**2).sum())
9     return float(np.dot(a,b)/d) if d>1e-12 else 0.0
10
11 def get_layer_profile(text, tokenizer, model):
12     """Returns [33, 5] correlation profile per layer."""
13     inp=tokenizer(text, return_tensors="pt",
14                   truncation=True, max_length=64).to(model.device)
15     with torch.no_grad():
16         out=model(**inp, output_hidden_states=True)
17     rows=[]
18     for hs in out.hidden_states:
19         v=hs[0].float().cpu().numpy()
20         hm=np.abs(v).mean(axis=0)
21         t_d=np.zeros(D); m=hm>1e-10
22         t_d[m]=np.log(hm[m])/np.log(PHI3)
23         rows.append([fast_corr(t_d, k)
24                     for k in [k6, k5, k4, k3, k2]])
25     return np.array(rows)

```

Listing 4: Layer-by-layer gate profile (Experiment 7).

F Two-Type Classification — Python

```

1 from scipy.stats import pearsonr, mannwhitneyu
2
3 # T5, H5: [N, 33, 5] from full run
4 T=T5[:,1:,:]; H=H5[:,1:,:] # [N, 32, 5]
5 delta=T-H
6 delta_mean=delta.mean(axis=0)
7
8 # Delta consistency at layer 29 (index 28)
9 dc=np.array([pearsonr(delta[i,28,:],
10                      delta_mean[28,:])[0] for i in range(N)])
11
12 recoverable=np.where(dc>0.5)[0] # n=16
13 fossil      =np.where(dc<0.0)[0] # n=16
14
15 # Cross-correlation at layer 20 (index 19)
16 cross_l20=np.array([pearsonr(T[i,19,:], H[i,19,:])[0]
17                     for i in range(N)])
18 stat,p=mannwhitneyu(cross_l20[recoverable],
19                    cross_l20[fossil],
20                    alternative='two-sided')
21 # Result: rec=0.697, fossil=0.936, p=0.0016

```

Listing 5: Fossil vs. recoverable (Experiment 13).